

M003 Extension — English Worksheet (Beginner)

Topic: FizzBuzz Pixel Art (remainders & rules) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 3) · **Time:** about 30 minutes

You already made pixel art by choosing each block's colour **by hand**. This time the **code** chooses every colour for you — using one rule, the **FizzBuzz rule**. Load the world, type **run** in the chat, and watch the agent paint a **15 × 15** wall one block at a time, counting from **1 to 225**.

If the block's number...	it says	colour
divides by 3 and 5	FIZZ BUZZ	● gold
divides by 3	FIZZ	■ black
divides by 5	BUZZ	■ purple
divides by neither	<i>(nothing)</i>	■ green

Big idea: the **number** on a block picks its colour. Divides by 3 → black. Divides by 5 → purple. Divides by 3 **and** 5 → gold. Nothing → green.

Here is one row (blocks 1 → 15). This same row repeats all the way up the wall.

🔑 **One row = blocks 1 → 15 (this row repeats up the whole wall)**

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

1 · Predict 🧠

Use the rule. Circle the colour.

Block 3 divides by 3. Circle its colour: gold · black · purple · green

Block 5 divides by 5. Circle its colour: gold · black · purple · green

Block 15 divides by 3 and 5. Circle its colour: gold · black · purple · green

Block 7 divides by nothing. Circle its colour: gold · black · purple · green

2 · Spot the Bug 🐛

Bug — Block 15 divides by 3 **and** 5, so it should be **gold**. In the broken code the $\div 3$ check runs first, so block 15 turns **black** by mistake.

```
if remainder of  $N \div 3 = 0$ :  
    paint BLACK    ← FIZZ  
else if remainder of  $N \div 3 = 0$  and remainder of  $N \div 5 = 0$ :  
    paint GOLD     ← FIZZ BUZZ
```

What colour should block 15 really be? Circle one: gold · black

Which check should go first? Circle one: $\div 3 \cdot \div 3$ and $\div 5$

3 · Show Your Work 📷 🎥

Load the world. Type run in the chat. Watch the agent count 1 → 225 and paint the wall.

 The finished wall — 15 × 15 (block 1 → 225)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
14	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
13	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
12	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
11	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
10	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
9	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
8	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
7	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
6	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
5	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
4	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
3	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
2	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold
1	green	green	black	green	purple	black	green	green	black	purple	green	black	green	green	gold

Every row is the same, so you see stripes going up.

How many gold blocks are in each row? Circle one: 1 · 2 · 3

Which colour is there the most of? Circle one: gold · black · purple · green

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the colours and the stripes.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.